Low-Level Design (LLD)

Configuration, rules, workflows and runbook

Document	Low-Level Design (LLD)
Project	SailPoint IdentityIQ - HR CSV authoritative source, lifecycle and LDAP ApacheDS provisioning
Version	1.1
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Source	github.com/ashok8033/Study - branch main, folder SailPointIIQ-HR-CSV

1. Environment

Item	Value
IIQ URL	http://localhost:8080/identityiq (Tomcat on Windows)
Source package	SailPointIIQ-HR-CSV/ (config/, data/, ldif/)
Master import file	config/init-HR-CSV.xml (run from the IIQ console: import .../config/init-HR-CSV.xml)
HR file	C:/SailPoint/hrfeed/HR_Employees.csv
ApacheDS	ldap://localhost:10389, bind uid=admin,ou=system / secret (lab default)
Logs	WEB-INF/classes/log4j2.properties -> C:/SailPoint/logs/sailpoint.log

2. HR Authoritative CSV application

Setting	Value
Connector	sailpoint.connector.DelimitedFileConnector (type DelimitedFile), authoritative
file / delimiter / hasHeader	C:/SailPoint/hrfeed/HR_Employees.csv / | / true
Account ID / display	employeeNumber
Rules	Correlation, Creation (IdentityCreation), Customization, Manager Correlation
Provisioning policy	Create Account Policy (fields from identity; date validation DD-MM-YYYY)

File layout

```
employeeNumber|firstName|lastName|startDate|endDate|managerId|jobDescription|country|employeeType|isManager
1000|Ashokkumar|Chidambaram|01-01-2014|||Managing Director|India|Employee|Y
1001|Rajesh|Kumar|01-04-2015||1000|Chief Executive Officer|India|Employee|N
```

Schema attribute	Notes
employeeNumber, firstName, lastName, startDate, endDate, managerId, jobDescription, country, employeeType, isManager	From the file
status, inactive, IIQDisabled	Calculated by HR CSV - Customization (Terminated -> IIQDisabled = true)

3. Identity attributes (ObjectConfig Identity)

Created by the setup rule **HR CSV - Setup Identity Mappings** (run once; safe to re-run). They are shown on View Identity by **HR CSV - Show Attributes On Identity Page** (UIConfig identityViewAttributes).

Identity attribute	Source	Target
employeeNumber, startDate, endDate, jobDescription, country, employeeType, isManager	HR CSV attribute of the same name	-
firstname / lastname / manager	HR firstName / lastName / managerId	-
inactive	Global rule Emergency Identity - Inactive Attribute (true if emergencyDisable, else HR inactive)	-
emergencyDisable	No source; editable (Permanent); set by the emergency workflow	-
displayName	Global rule HR CSV - Display Name	LDAP ApacheDS: displayName
status	Global rule HR Lifecycle - Status Attribute	-
email	LDAP ApacheDS: mail (generated by the Create Account policy)	-

Status calculation (whole days; still Active on the end date):

```
if emergencyDisable -> Disabled   (overrides the dates)
if endDate  < today  -> Terminated
if startDate > today -> PreHire
otherwise            -> Active
```

4. Rules

Rule	Type	Purpose
HR CSV - Customization	ResourceObjectCustomization	Trim values, validate dates, normalise isManager Y/N, set status / inactive / IIQDisabled
HR CSV - Correlation	Correlation	identityName = employeeNumber
HR CSV - Manager Correlation	ManagerCorrelation	Manager identity name = managerId
HR CSV - Identity Creation	IdentityCreation	Name = employeeNumber, first/last/display name, type
HR Lifecycle - Status Attribute	IdentityAttribute (global)	Disabled if emergencyDisable, else PreHire / Active / Terminated from HR dates
HR CSV - Display Name	IdentityAttribute (global)	firstName + ' ' + lastName
LDAP ApacheDS - Correlation	Correlation	identityName = LDAP employeeNumber
LDAP ApacheDS - Library	Library	JNDI connect, group add/remove, disable/enable, read attributes, time format
LDAP ApacheDS - Before Provisioning	BeforeProvisioning	Remove 'groups' from the request; keep changes as hrGroupsAdd / hrGroupsRemove
LDAP ApacheDS - After Provisioning	AfterProvisioning	Apply group changes as uniqueMember add/remove
LDAP ApacheDS - Customization	ResourceObjectCustomization	Read groups and account status (pwdAccountLockedTime etc.)
LDAP ApacheDS - Set Account Status	Called rule	Disable / enable LDAP account and update the IIQ link
LDAP ApacheDS - Disable Terminated Users	Utility	Catch-up: disable LDAP for identities already Terminated
HR CSV - Setup Identity Mappings	Setup	Create identity attributes, sources, global rules and LDAP target
HR CSV - Show Attributes On Identity Page	Setup	Add attributes to UIConfig identityViewAttributes
HR CSV - Diagnose	Utility (read-only)	Mappings (DB / cache), HR link, identity values, status rule result
Aggregate Single Account	Utility	Targeted aggregation of one account for any application
HR Lifecycle - Send Test Email	Utility	Show mail settings and send one test email
Emergency Identity - Inactive Attribute	IdentityAttribute (global)	inactive = emergencyDisable OR HR inactive
Emergency Identity - Provision Accounts	Called rule	Disable / enable every non-authoritative account; returns status + per-account details
Emergency Identity - Audit	Called rule	Write the AuditEvent and the log line
Emergency Identity - Setup	Setup	Create emergencyDisable, switch inactive to the global rule, add audit actions

5. LDAP ApacheDS application

Setting	Value
Connector	sailpoint.connector.LDAPConnector (type LDAP)
host / port / SSL	localhost / 10389 / false; authorizationType simple
Account search DN / filter	ou=people,dc=example,dc=com / (objectClass=inetOrgPerson), SUBTREE
Group search DN / filter	ou=groups,dc=example,dc=com / (objectClass=groupOfUniqueNames); member attribute uniqueMember
beforeProvisioningRule / afterProvisioningRule	LDAP ApacheDS - Before Provisioning / After Provisioning (application attributes)
CustomizationRule / CorrelationRule	LDAP ApacheDS - Customization / Correlation

Account schema (identityAttribute dn, display uid): uid, cn, sn, givenName, displayName, mail, employeeNumber, employeeType, title, objectClass (multi), groups (entitlement, managed, multi), accountStatus, pwdAccountLockedTime, lockedTime, lastFailedLogin, passwordLastChanged, createdOn, modifiedOn.

Create Account Policy

Field	Value
uid	firstname.lastname (lower case), made unique with 2, 3, ... against existing LDAP links
dn	uid=<uid>,ou=people,dc=example,dc=com
objectClass	top, person, organizationalPerson, inetOrgPerson
cn / displayName / sn / givenName	identity display name / display name / last name / first name
mail / employeeNumber / title / employeeType	<uid>@example.com / identity name / jobDescription / employeeType
password	Welcome@<employeeNumber> (lab only)

Account status mapping

pwdAccountLockedTime	accountStatus	IIQ flags
not present	Active	-
000001010000Z (admin lock)	Disabled	IIQDisabled = true
any other timestamp (failed logins)	Locked	IIQLocked = true; lockedTime set

6. ApacheDS directory layout (LDIF)

```
dc=example,dc=com
  ou=people   users created by IIQ: uid=<first.last>
  ou=groups   groupOfUniqueNames (placeholder member uid=admin,ou=system)
              cn=AllEmployees, cn=Contractors, cn=Managers, cn=Office365Access
```

Files: ldif/ApacheDS-ou-people-and-groups.ldif, ldif/ApacheDS-group-Office365Access.ldif. The dc=example,dc=com partition must exist (Studio: server configuration > Partitions).

7. Roles and entitlements

Role	Type	Assignment / requirement	Entitlement
Birthright - All Active Users	business	MatchTerm status = Active	requires LDAP - All Employees
Birthright - Contractors	business	status = Active AND employeeType = Contractor	requires LDAP - Contractors
Birthright - Managers	business	status = Active AND isManager = Y	requires LDAP - Managers
Office365 Access	business	No selector - requestable only	requires LDAP - Office365Access
LDAP - All Employees / Contractors / Managers / Office365Access	it	Profile on LDAP ApacheDS	groups CONTAINS_ALL cn=...,ou=groups,dc=example,dc=com

ManagedAttribute: groups = cn=Office365Access,ou=groups,dc=example,dc=com (requestable, with description).

8. Lifecycle event, workflow and email

Object	Configuration
IdentityTrigger 'HR Lifecycle - Status Change'	type AttributeChange, attributeName status, handler WorkflowTriggerHandler, HandlerParameters/Attributes: workflow = HR Lifecycle - Status Change Notification
Workflow 'HR Lifecycle - Status Change Notification'	Collect Details -> [Disable LDAP | Enable LDAP] -> Check Notify -> Send Email -> Stop
EmailTemplate 'HR Lifecycle - Status Change'	Name, employee number, old / new status, dates, job, manager, LDAP result
Recipient	notifyEmail = ashokhearts@gmail.com

```
Collect Details : oldStatus / newStatus from the IdentityChangeEvent
Disable LDAP    : newStatus = Terminated and oldStatus != Terminated
                  -> rule LDAP ApacheDS - Set Account Status (action Disable)
Enable LDAP     : oldStatus = Terminated and newStatus = Active
                  and emergencyDisable is not true
                  -> rule LDAP ApacheDS - Set Account Status (action Enable)
Check Notify    : email only when oldStatus and newStatus are set and differ
```

9. Emergency Identity Disable / Enable

Object	Configuration
SPRight / Capability	EmergencyIdentityDisableEnable / EmergencyIdentityAdministrator
DynamicScope 'Emergency Identity Administrators'	capabilities.name = SystemAdministrator OR EmergencyIdentityAdministrator
QuickLink 'Emergency Identity Disable / Enable'	action workflow, category Manage, workflowName = Emergency Identity Status Management Workflow
Workflow 'Emergency Identity Status Management Workflow'	transient; no approvals; form owner = launcher
File	config/QuickLink/QuickLink-Emergency_Identity-DisableEnable.xml, config/Workflow/Workflow-Emergency_Identity-StatusManagement.xml

```
Request Form      : targetIdentity (Identity, postBack), read-only Identity Name, Current Status,
                    Emergency Disable, HR Status; requestedAction (only the valid value);
                    justification (required). Cancel -> Stop
Validate Request  : launcher has SystemAdministrator / EmergencyIdentityAdministrator; target exists,
                    is not the launcher; action matches emergencyDisable -> else DENIED -> Audit
Update Identity   : ObjectUtil.lockIdentity; emergencyDisable = (action == Disable);
                    status = rule HR Lifecycle - Status Attribute (Disabled / HR value);
                    inactive = rule Emergency Identity - Inactive Attribute; unlock (commits)
Provision Accounts: rule Emergency Identity - Provision Accounts -> provisioningResult
Collect Result    : workflowResult = SUCCESS | PARTIAL | FAILED | NOTHING_TO_DO
Audit             : rule Emergency Identity - Audit
```

```
Show Result       : read-only form for the requestor -> Stop
```

Account handling (Emergency Identity - Provision Accounts)

Account	Disable	Enable
Authoritative (HR Authoritative CSV)	skipped	skipped
Already in the requested state	skipped	skipped
Any account, HR status Terminated	processed	skipped (stays disabled)
LDAP ApacheDS	rule LDAP ApacheDS - Set Account Status (Disable)	same rule (Enable)
Connector supports ENABLE	AccountRequest Disable via Provisioner	AccountRequest Enable via Provisioner
Other	MANUAL (counted as failed)	MANUAL (counted as failed)

Audit event

Field	Value
action	EmergencyIdentityDisable / EmergencyIdentityEnable / EmergencyIdentityDenied
source / target / created	requestor / target identity / date and time
string1	workflow result: SUCCESS, PARTIAL, FAILED, NOTHING_TO_DO, DENIED
string2 / string3 / string4	justification / emergencyDisable before -> after / account summary
attributes	requestedAction, accounts (full list), workflowCaseId

10. Tasks

Task	Type	Key options
HR Authoritative CSV - Account Aggregation	AccountAggregation	applications = HR Authoritative CSV, checkDeleted, noOptimizeReaggregation
HR Lifecycle - Daily Identity Refresh	Identity refresh	promoteAttributes, processTriggers, correlateEntitlements, provision, synchronizeAttributes, refreshManagerStatus
LDAP ApacheDS - Account Aggregation	AccountAggregation	applications = LDAP ApacheDS, checkDeleted

Recommended schedule: HR aggregation -> Daily Identity Refresh (00:05) -> LDAP aggregation.

11. Deployment runbook

Step	Action
1	ApacheDS: create the dc=example,dc=com partition; import both LDIF files; enable Password Policy
2	Copy data/HR_Employees.csv to C:/SailPoint/hrfeed/
3	IIQ console: import .../config/init-HR-CSV.xml
4	Applications > LDAP ApacheDS: re-enter bind password, Test Connection; check the Rules tab
5	rule "HR CSV - Setup Identity Mappings" then rule "HR CSV - Show Attributes On Identity Page"; log out / in
6	Global Settings > Mail Settings: smtp.gmail.com, 587, TLS, app password; rule "HR Lifecycle - Send Test Email"
7	Run HR aggregation, Daily Identity Refresh, LDAP aggregation
8	rule "LDAP ApacheDS - Disable Terminated Users" once for existing leavers
9	Schedule the three tasks
10	rule "Emergency Identity - Setup" (after step 5); give non-admin operators the Emergency Identity Administrator capability
11	Run the Daily Identity Refresh once so inactive is recalculated by the new global rule

12. Test cases

#	Scenario	Steps	Expected result
T1	Initial load	HR aggregation + refresh	16 identities named 1000-1015; Ashok is everyone's manager
T2	Status	Diagnose rule for 1000	status = Active; 1009/1011/1015 Terminated; 1014 PreHire
T3	Joiner	Set 1014 startDate to today; aggregate + refresh	Status Active; LDAP uid=james.taylor created in cn=AllEmployees; email
T4	Leaver	Set endDate to yesterday; aggregate + refresh	Status Terminated; HR account disabled; LDAP pwdAccountLockedTime = 000001010000Z; groups removed; email
T5	Rehire	Clear endDate; aggregate + refresh	Status Active; LDAP lock removed; email
T6	Name change	Change lastName; aggregate + refresh	Identity last / display name updated; LDAP displayName updated
T7	Request	Request Office365 Access; approve	User in cn=Office365Access
T8	Status read-back	Lock an entry in Studio; LDAP aggregation	accountStatus Disabled / Locked in IIQ
T9	Emergency access	Log in as a user without the capabilities	Quick link not shown
T10	Emergency disable	spadmin: quick link, active user, Disable	Only Disable offered; emergencyDisable = true; inactive; LDAP locked; SUCCESS; audit EmergencyIdentityDisable
T11	Emergency enable	Same user again, Enable	Only Enable offered; emergencyDisable = false; LDAP unlocked; audit EmergencyIdentityEnable
T12	Refresh keeps state	Emergency-disable a user; run Daily Identity Refresh	Status stays Disabled whatever the HR dates; identity inactive; LDAP stays locked; birthright groups removed

13. Troubleshooting

Symptom	Cause / fix
Import: 'must match DOCTYPE root null'	File has no DOCTYPE - every file in the package now has one
Unable to open filestream	File path wrong or under OneDrive; use C:/SailPoint/hrfeed and check with Test-Path
could not resolve property: employeeNumber	Non-searchable attribute; correlation now uses identity name
Attributes missing on identity	Run the setup rule, then the Daily Identity Refresh; check with HR CSV - Diagnose
LDAP NO_SUCH_ATTRIBUTE groups	Handled by Before / After Provisioning rules
Account not updated after HR change	IIQ reads a different file; check File Path, schema Preview, Aggregate Single Account output
pwdAccountLockedTime not visible in Studio	Enable 'Fetch operational attributes' in the connection's Browser Options
No email	Mail settings / Gmail app password; run HR Lifecycle - Send Test Email
Emergency quick link not shown	User lacks SystemAdministrator / EmergencyIdentityAdministrator; log out and in after granting
Emergency result PARTIAL / MANUAL	An application has no ENABLE feature or provisioning failed; see the result lines and the audit event
Refresh re-activates an emergency-disabled identity	inactive still mapped from HR; run rule "Emergency Identity - Setup"
Java heap space	setenv.bat CATALINA_OPTS -Xms1024m -Xmx4096m, or tomcat9w.exe Java tab for the service